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PI Name:	Gaver, Donald Paul Ph.D.		
Project Title:	Investigations of the Influence of Air-liquid Interfacial Stresses on Pulmonary Epithelial Cells in a Microgravity Environment		
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PI Email:	donald.gaver@tulane.edu	**Fax:**	FY (504) 862-8779
PI Organization Type:	UNIVERSITY	**Phone:**	(504) 865-5150
Organization Name:	Tulane University		
PI Address 1:	Biomedical Engineering		
PI Address 2:			
PI Web Page:			
City:	New Orleans	**State:**	LA
Zip Code:	70118	**Congressional District:**	2
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Contact Monitor:	Balasubramaniam, R.	**Contact Phone:**	216-433-2878
Contact Email:	RBalasubramaniam@grc.nasa.gov		
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	Pulmonary airway closure is a potentially dangerous event that can occur in microgravity environments and may result in limited gas exchange for flight crew during long-term space flight. To reopen collapsed airways, a semi-infinite bubble of air must separate airway walls that are held in apposition by a viscous lining fluid. However, the fluid mechanical stresses exerted by the migrating bubble can be very large and damage the lung. The fluid mechanics issues related to these studies are complex and are intimately related to the physiological responses of the lung. We will investigate the cellular responses to interfacial and viscous stresses that occur as a surfactant-doped bubble progresses over pulmonary epithelial cells that line the airways and alveoli in the lung. By identifying the magnitude of these stresses and the physiological response to the stimulus, we will provide insight of appropriate methods for opening collapsed airways, and how these methods depend upon the physicochemical properties of the airway lining fluid. As such, our studies will develop the link between fluid physics and potentially hazardous life-science issues that may be manifested during long-term space flight. Specific Aim #1: Study the fluid dynamical and surfactant physicochemical responses related to the opening of collapsed pulmonary airways in a microgravity environment. Specific Aim #2: Examine cellular trauma due to bubble progression in a physical model of a collapsed airway. Specific Aim #3: Predict flow and surfactant behavior that will minimize cellular injury during the opening of collapsed airways.
Task Description:	Our studies are of importance to NASA for two reasons. First, pulmonary airway closure is more ubiquitous in a microgravity environment than on earth - this has been demonstrated in microgravity life-science studies of pulmonary function during short-term parabolic flights and long-term investigations on the SpaceLab (SLS-1). In microgravity, airway closure exists throughout the lung, rather than only in the basal 'gravitationally dependent'regions. While this closure occurs primarily at low lung volumes, recent parabolic flight studies of inhaled gas boluses indicate that airway closure in microgravity occurs in close proximity to airways that remain open, and may provide the inhomogeneity in the lung. Such closure may impact pulmonary function in microgravity environments during 'normal' and disease conditions. It is thus important to understand the complex fluid-structure interaction of this system to determine the likelihood of pulmonary cellular damage in microgravity,and to identify methods of reducing the effects of these stresses on pulmonary cells. In addition, airway closure occurs in disease, and thus further understanding of the interaction between air-liquid interface behavior and pulmonary cellular responses could lead to advanced therapeutics for individuals on earth. The small physical size of the airway/bubble system that exists in regions of airway closure means that gravitational effects are small. While flight studies of this system are not proposed herein, we may in the future find it important to scale-up this system so as to measure the convection, transport and stress behavior in a system of larger physical size. Such studies would have to be performed in a microgravity environment to remove the effects of gravity. However, it is important to develop a basic understanding of these systems in a terrestrial environment so that we can determine the scaling principles from which the microgravity flight studies would be based.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The overall goal of this project has been to understand and suggest methods for treating pathological behavior of the lung in microgravity environments. In particular, we have sought to understand the fluid/structure and physicochemical interactions that occur between delicate pulmonary tissue and the lining fluid that coats the interior surfaces of the lung. In microgravity (as well as on earth) this liquid lining can become unstable and occlude airways, thus reducing the lung’s ability to transport O2 and CO2.
Task Progress:	Investigation I: Experimental investigations of the effect of asymmetric oscillation flow conditions on a airway reopening and surfactant transport of atelectic pulmonary airways In patients with acute respiratory distress syndrome (ARDS), a reduction in ventilation to the affected airways occurs and ventilator-induced lung injury often results during the process of airway reopening. Our NASA-funded studies have established that this damage originates from large mechanical stress gradients, primarily normal-stress gradients, that develop as air is forced through occluded airways. Understanding the physical principles and behavior of dynamic air-liquid interfaces is vital to reducing the occurrence and intensity of lung injury associated with ARDS ventilation treatments. In the present experimental study, a model of an airway, consisting of a narrow-bore (1 mm ID) glass tube maintained at 37oC, is utilized to investigate the effects of an asymmetric oscillation on surfactant transport to the air-liquid interface. We hypothesize that an asymmetric oscillation will increase surfactant transport to the interface by building up a multilayer of surfactant at the bubble tip that can remobilize the interface. In our experimental apparatus, the proximal end of the tube serves as a reservoir of liquid, while an oscillating finger of air is slowly progressed through the tube. This research was published in the Biophysical Journal. Results from the experiments indicated that a slow reverse asymmetric oscillation of the interface affects the surfactant transport noticeably in comparison to both a fast reverse asymmetric oscillation and a symmetric oscillation. The capillary pressure decreases considerably as the interface moves forward in a slow reverse asymmetry. We propose that this behavior can be attributed to a surfactant build up during the reverse flow that acts as a reservoir during the fast forward motion of the interface. The combination of a large surfactant reservoir and fast forward flow may allow for successful surfactant transport and adsorption to the expanding interface, thus, lowering surface pressures. In the absence of surfactant (water alone), fast reverse and fast forward flow conditions have similar pressures with diminished hysteresis, indicating that a fast reverse asymmetry affects surfactant behavior at the interface. The decrease in pressures seen in fast reverse asymmetric conditions will lead to decreases in mechanical stress in occluded airways and ultimately a decrease in ventilator induced lung injury. Investigation II: Computational predictions of interfacial behavior during rapid oscillation of a finger of air in a cylindrical channel. In this study we computationally investigate the unsteady propagation of a finger of air through a liquid-filled rigid tube. The system, a free surface with fluid-structure interaction, is modeled using the boundary element method (BEM)

coupled to lubrication theory in the upstream thin film region. Our axisymmetric model consists of a BEM region spanning from the downstream end to a point on the air/liquid interface where small slope assumptions hold. Further upstream, lubrication theory is used to resolve the thin film. This study demonstrated the modification of stress with variation in fluid flow characteristics (pulsatility). This research was published in the Journal of Fluid Mechanics.

Investigation III: In vitro Investigation of the Effects of Airway Reopening on Pulmonary Epithelial Topographic and Barrier Function Integrity

During mechanical ventilation, the pulmonary epithelium is exposed to pathological levels of mechanical stress as the ventilating air front forces its way through the fluid that obstructs the airways. Prior studies have shown that the normal-stress gradient causes damage to airway epithelium; but, the mechanism of damage has not yet been established.

Our prior studies have indicated that slower bubble progression in the flow chamber-model of airway reopening causes considerable epithelial cell injury and/or death. The present studies reinforce these previous observations; in addition, they demonstrate a potential injury mechanism that may promote the development of pulmonary edema during mechanical ventilation. Experimentally, we find that the tight junction protein ZO-1 tends to dissociate from the cell periphery into the cytoplasm when the epithelium is subjected to the passage of air through the flow chamber. In an epithelial cell layer, this would have the effect of diminishing the strength the the paracellular tight junctions and increasing the leakiness of the epithelium to fluid and proteins.

The stresses that arise during airway reopening compress the epithelium and to a lesser degree pull tangentially along the cell surface. Our computational model shows that these stresses are transmitted through the epithelium to generate significant forces at the focal adhesion contact sites between the cell layer and its underlying extracellular matrix, which could, in turn, influence cell mechanotransduction. These basolateral forces are most excessive in the trasepithelial cell regions where tight junctions concentrate. In addition, they are magnified at slower reopening velocities where experimentally observed cell damage and ZO-1 disruption is most prominent.

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